ISSN 1607-6729, Doklady Biochemistry and Biophysics, 2024, Vol. 516, pp. 111–114. © Pleiades Publishing, Ltd., 2024.
ISSN 1607-6729, Doklady Biochemistry and Biophysics, 2024. © Pleiades Publishing, Ltd., 2024.

BIOCHEMISTRY, BIOPHYSICS, AND MOLECULAR BIOLOGY

Binary Proton Therapy of Ehrlich Carcinoma Using Targeted Gold Nanoparticles

M. V. Filimonova[a,b], D. D. Kolmanovich[c,d], G. V. Tikhonowski[e], D. S. Petrunya[d,e,*], P. A. Kotelnikova[f], A. A. Shitova[a], O. V. Soldatova[a], A. S. Filimonov[a], V. A. Rybachuk[a], A. O. Kosachenko[a], K. A. Nikolaev[a], G. A. Demyashkin[a], A. A. Popov[e], M. S. Savinov[e], A. L. Popov[c,d], I. V. Zelepukin[f], A. A. Lipengolts[e,g], K. E. Shpakova[e,g], A. V. Kabashin[h], S. N. Koryakin[a,b], Academician S. M. Deyev[f], and I. N. Zavestovskaya[d,e]

Received January 25, 2024; revised February 15, 2024; accepted February 20, 2024

Abstract—Proton therapy can treat tumors located in radiation-sensitive tissues. This article demonstrates the possibility of enhancing the proton therapy with targeted gold nanoparticles that selectively recognize tumor cells. Au-PEG nanoparticles at concentrations above 25 mg/L and 4 Gy proton dose caused complete death of EMT6/P cells in vitro. Binary proton therapy using targeted Au-PEG-FA nanoparticles caused an 80% tumor growth inhibition effect in vivo. The use of targeted gold nanoparticles is promising for enhancing the proton irradiation effect on tumor cells and requires further research to increase the therapeutic index of the approach.

Keywords: proton therapy, radiosensitizers, nanoparticles, gold, tumor therapy

DOI: 10.1134/S1607672924700819

INTRODUCTION

Proton radiation therapy is a promising direction in radiation oncology. The main advantages of proton therapy are the release of most of the energy of protons at the end of their path in the Bragg peak and the straightness of the particle trajectory in human tissues [1]. This favorably distinguishes this approach from photon radiation therapy, in which the maximum radioactive dose is absorbed at the beginning of pathway into tissues. Binary approaches to proton therapy with the use of nanoparticles as sensitizers can increase toxic effects through the physical effects of dose enhancement as well as the biological effects of the generation of reactive oxygen species and radicals [2, 3]. In particular, proton irradiation of nanoparticles of high atomic number (Z) elements such as Au and Pt was shown to cause the release of secondary and Auger electrons and the formation of hydroxyl radicals in solution [4]. On the other hand, even light elements such as boron [5] and titanium [6] were able to enhance proton therapy through catalytic processes occurring on the surface of particles.

However, the delivery of nanoparticles to tumor tissue before radiation therapy is an independent complex task. Actively dividing tumor cells in most cases are located at the periphery of the tumor, as well as near blood vessels. Because of this, intratumoral administration of particles is ineffective, since their diffusion is limited by binding to the extracellular matrix [7]. When using intravenous administration, the presence of targeting vectors that can selectively recognize the periphery of tumor tissue and, upon binding, retain nanoparticles in the target tissue for a long time is required [8]. Modern studies on the creation of targeted particles for proton therapy are limited to the in vitro level [9, 10], although they show high selectivity of the effect.

[a] *Tsyb Medical Radiological Research Center—Branch of the National Medical Research Radiological Centre of the Ministry of Health of the Russian Federation, Obninsk, Russia*

[b] *Obninsk Institute for Nuclear Power Engineering—Branch of the National Research Nuclear University MEPhI, Obninsk, Russia*

[c] *Institute of Theoretical and Experimental Biophysics, Russian Academy of Sciences, Pushchino, Russia*

[d] *Lebedev Physical Institute, Russian Academy of Sciences, Moscow, Russia*

[e] *National Research Nuclear University MEPhI (Moscow Engineering Physics Institute), Moscow, Russia*

[f] *Shemyakin–Ovchinnikov Institute of Bioorganic Chemistry, Russian Academy of Sciences, Moscow, Russia*

[g] *Blokhin National Medical Research Center of Oncology of the Ministry of Health of the Russian Federation, Moscow, Russia*

[h] *Aix-Marseille University, CNRS, Marseille, France*

**e-mail: d.petrunya@lebedev.ru*

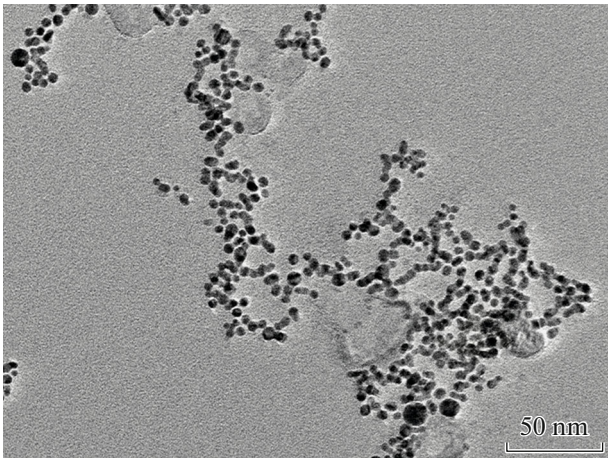

Fig. 1. Transmission electron microscopy images of Au nanoparticles.

A promising target for targeting is the FRα isoform of the folate receptor. This protein is present in a large number of cancers of epithelial origin, including breast, lung, kidney, and ovarian cancers [11] with an expression level of 1–10 million proteins per cell, which is 100–300 times higher than in healthy tissues. In this work, we propose a fundamental approach to binary proton therapy using targeted gold nanoparticles targeting the folate receptor using albumin conjugated to folic acid.

RESULTS AND DISCUSSION

To demonstrate the described approach, elemental gold nanoparticles were obtained by pulsed laser ablation [12] in a 1 mM aqueous NaCl solution. Irradiation of a gold target using a femtosecond Yb:KGW laser (TETA-10, wavelength 1030 nm, pulse duration 270 fs, pulse energy 50 μJ, frequency 200 kHz) resulted in the formation of nanoparticles with a size of less than 100 nm. For in vitro and in vivo experiments, nanoparticles with a size less than 20 nm were isolated by centrifugation. As a result, spherical particles with a size of 4 to 8 nm were obtained, which was confirmed by transmission electron microscopy (Fig. 1). Thus obtained nanoparticles were coated with polyethylene glycol (5 kDa) with a terminal thiol group due to the formation of a covalent Au–S bond [13], which ensured their long-term colloidal stability in phosphate-buffered saline for at least 24 h.

To evaluate the possibility of enhancing proton therapy using the obtained nanoparticles, the cytotoxicity of the binary approach was studied on mouse mammary carcinoma cells (EMT6/P line). Gold nanoparticles were introduced to the cells and coincubated with them for 16 h, after which the cells were irradiated at the Prometheus proton therapy complex (Protvino, Russia). The proton energy in the beam was 160.5 MeV, and the cell monolayer was located at the position of the distributed Bragg peak. The beam homogeneity was at least 98% at an isodose level of 95%. An assessment of the number of colonies with a size of more than 50 cells 8 days after irradiation showed a significant decrease in clonogenic activity with binary therapy (Fig. 2).

Importantly, Au-PEG nanoparticles in the concentration range studied (10–50 μg/mL) did not cause a decrease in the number of formed cell colonies, which confirms their high biocompatibility. At the same time, the cells showed a dose-dependent response to irradiation with a proton beam, which was expressed in a decrease in the number of the formed colonies by 25 and 50% for doses of 2 and 4 Gy, respectively. Preincubation of cells with Au-PEG

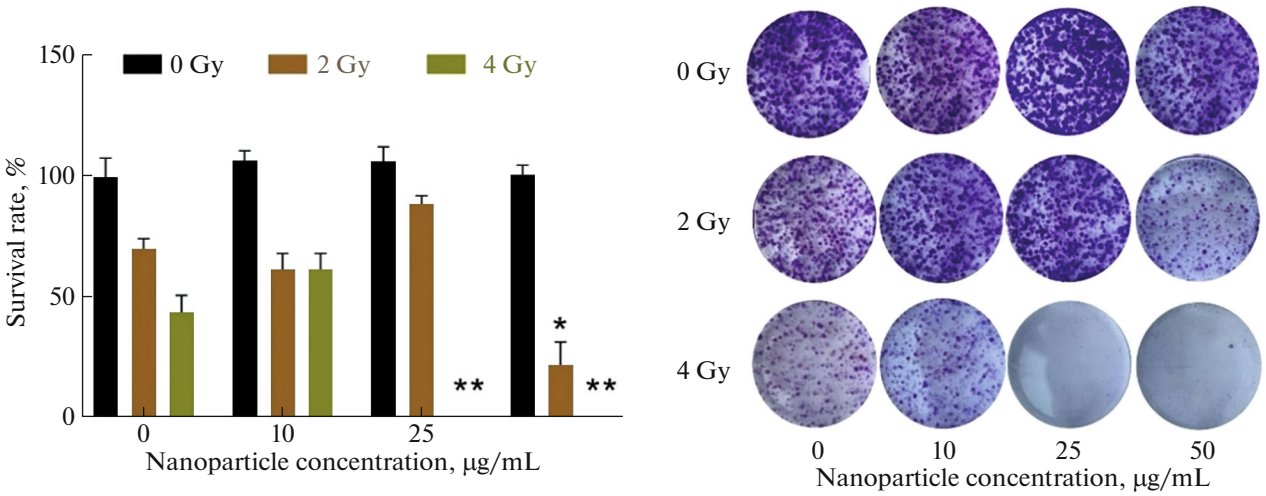

Fig. 2. Clonogenic analysis of EMT6/P adenocarcinoma cells after proton beam irradiation in the presence of Au-PEG nanoparticles; ** $p < 0.001$, * $p < 0.05$, Student's t test.

nanoparticles at a concentration of 50 μg/mL with co-irradiation at a dose of 2 Gy resulted in a fourfold decrease in the number of colonies compared to the non-irradiated group (Fig. 2). In addition, preincubation of cells with nanoparticles at concentrations of 25 and 50 μg/mL and irradiation at a dose of 4 Gy led to the complete disappearance of cancer colonies. Thus, Au-PEG nanoparticles exhibited a high biocompatibility and a radiosensitizing effect when irradiated with a proton beam at a dose of 2–4 Gy in vitro.

The radiosensitizing properties of gold were then studied in vivo. Solid Ehrlich carcinoma was chosen as a tumor model. These cells overexpress folate receptor alpha [14], so it was possible to target intravenously administered nanoparticles using a folic acid-based targeting module. To demonstrate this principle, we prepared conjugates of Au-PEG nanoparticles with folic acid using bovine serum albumin (BSA) as an intermediate linker (hereinafter, Au-PEG-FA). The conjugate with BSA was obtained by carbodiimide chemistry. The attachment of the conjugate to the nanoparticles was determined by the formation of a bond between the SH groups in albumin and the free surface of gold NPs. The effectiveness of attachment of folic acid to nanoparticles was confirmed using UV–Vis optical spectroscopy by the intensity of the characteristic absorption peak of folic acid (363 nm) [15].

Ehrlich carcinoma cells were injected into female F1(CBA×C57BL6j) mice (age 8–10 weeks, body weight 22–24 g) subcutaneously in the area of the lateral surface of the right femur. Ten days after the injection, the right femur was irradiated with a proton beam using the Prometheus complex with a radiation dose of 31 Gy. To evaluate binary therapy, one group of mice was injected intravenously with 20 mg Au-PEG-FA nanoparticles with folic acid on the surface 24 h before irradiation in 0.2 mL PBS. The accumulation of nanoparticles in the tumor node was assessed using computed tomography. The criterion for the presence of nanoparticles in a certain area was an increase in radiodensity compared to native sections (in soft tissue before the introduction of nanoparticles, ~110–130 HU). After intravenous injection of nanoparticles, CT data showed an inhomogeneous increase in the radiodensity of the tumor tissue to ~210–240 HU in the form of small multiple hyperdense inclusions distributed unevenly throughout the entire tumor volume. The most intense contrast was observed in the peritumoral stroma of the tumor node, where the largest number of living tumor cells is expected to be located. These data indicate that the folic acid conjugate is an effective tumor-specific vector for solid Ehrlich carcinoma.

The dynamics of the development of Ehrlich carcinoma in the experimental groups was assessed morphometrically. For this purpose, every 2–3 days of observation, the linear dimensions of tumor nodes were measured in all animals using a ShTsTs-1-125 digital caliper (NPP ChIZ, Russian Federation). The volume of tumor nodes and the tumor growth inhibition (TGI) index at the stages of observation were calculated according as described in [16].

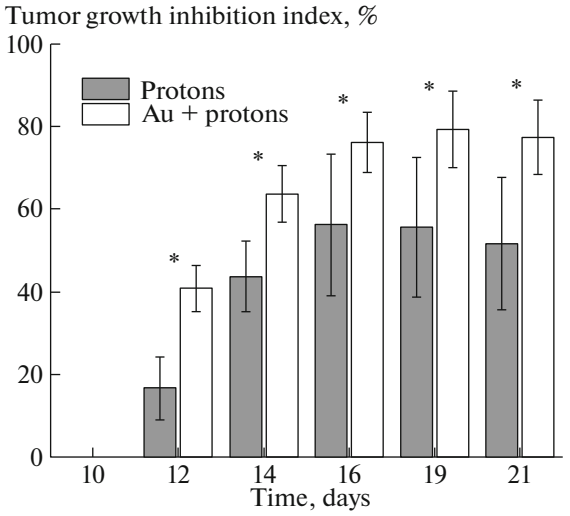

Fig. 3. Tumor growth inhibition (TGI) index values for Ehrlich carcinoma after exposure to protons at a dose of 31 Gy in the presence and absence of Au-PEG-FA nanoparticles; * $p < 0.05$, Kruskal–Wallis test.

The use of Au-PEG-FA nanoparticles as a radiosensitizer during proton beam therapy significantly enhanced the biological response of Ehrlich carcinoma in vivo. After a single local irradiation with protons at a dose of 31 Gy, a pronounced inhibition of tumor growth was observed (TGI = 55–60%). At the same time, preliminary intravenous injection of nanoparticles statistically significantly (by 15–25%) increased tumor growth inhibition throughout the entire observation period, with the TGI value reaching 80% (Fig. 3). It should be noted that no toxic (lethal) effects of proton and binary therapy in the experiment were observed.

The effect of radiosensitization was also confirmed by a comparative analysis of the weight of tumor nodes isolated at the terminal stage of the experiment. In mice after binary therapy, the tumor weight was 2.2 times lower compared to the mice without therapy and 1.25 times lower compared to the mice that received only proton therapy. In this study, all animals were euthanized for terminal macroscopic examination of tumor nodes.

Thus, in this study, we showed that targeted gold nanoparticles targeting the tumor folate receptor are promising proton therapy sensitizers. The nanoparticles caused a complete inhibition of the clonogenic activity of EMT6/P adenocarcinoma in vitro at concentrations above 25 μg/mL and a radiodose of 4 Gy. In addition, preliminary results of therapy showed a significant additional deceleration of tumor growth at binary exposure to nanoparticles and protons. Binary

therapy using targeted gold nanoparticles requires further study to determine the toxicity of the approach, detailed particle biodistribution, and long-term tumor treatment effects.

FUNDING

The study was supported by the Ministry of Science and Higher Education of the Russian Federation (project no. 075-15-2021-1347).

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

Protocols using animals were approved by the Commission for Bioethical Control over the Keeping and Use of Laboratory Animals for Scientific Purposes of the National Medical Research Center of Radiology of the Ministry of Health of the Russian Federation (Application no. 1-N-00053, dated September 12, 2023).

CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

REFERENCES

1. Durante, M. and Loeffler, J.S., *Nat. Rev. Clin. Oncol.*, 2010, vol. 7, no. 1, pp. 37–43.
2. Lo, C.Y., Tsai, S.W., Niu, H., Chen, F.H., Hwang, H.C., Chao, T.C., et al., *ACS Omega*, 2023, vol. 8, no. 20, pp. 17922–17931.
3. Martínez-Rovira, I. and Prezado, Y., *Med. Phys.*, 2015, vol. 42, no. 11, pp. 6703–6710.
4. Zwiehoff, S., Johny, J., Behrends, C., Landmann, A., Mentzel, F., Baumer, C., et al., *Small*, 2022, vol. 18, no. 9, p. 2106383.
5. Zavestovskaya, I.N., Popov, A.L., Kolmanovich, D.D., Tikhonowski, G.V., Pastukhov, A.I., Savinov, M.S., et al., *Nanomaterials*, 2023, vol. 13, no. 15, p. 2167.
6. Gerken, L.R., Gogos, A., Starsich, F.H., David, H., Gerdes, M.E., Schiefer, H., et al., *Nat. Commun.*, 2022, vol. 13, no. 1, p. 3248.
7. Zelepukin, I.V., Griaznova, O.Y., Shevchenko, K.G., Ivanov, A.V., Baidyuk, E.V., Serejnikova, N.B., et al., *Nat. Commun.*, 2022, vol. 13, no. 1, p. 6910.
8. Tolmachev, V.M., Chernov, V.I., and Deyev, S.M., *Russ. Chem. Rev.*, 2022, vol. 91, no. 3, p. RCR5034.
9. Li, S., Bouchy, S., Penninckx, S., Marega, R., Fichera, O., Gallez, B., et al., *Nanomedicine*, 2019, vol. 14, no. 3, pp. 317–333.
10. Kang, S.H., Hong, S.P., and Kang, B.S., *Int. J. Radiat. Biol.*, 2018, vol. 94, no. 11, pp. 1006–1016.
11. Siwowska, K., Haller, S., Bortoli, F., Benesova, M., Groehn, V., Bernhardt, P., et al., *Mol. Pharm.*, 2017, vol. 14, no. 2, pp. 523–532.
12. Popov, A.A., Swiatkowska-Warkocka, Z., Marszalek, M., Tselikov, G., Zelepukin, I.V., Al-Kattan, A., et al., *Nanomaterials*, 2022, vol. 12, no. 4, p. 649.
13. Gao, J., Huang, X., Liu, H., Zan, F., and Ren, J., *Langmuir*, 2012, vol. 28, no. 9, pp. 4464–4471.
14. Shiji, R., Joseph, M.M., Sen, A., Unnikrishnan, B.S., and Sreelekha, T.T., *Int. J. Biol. Macromol.*, 2022, vol. 219, pp. 740–753.
15. Baibarac, M., Smaranda, I., Nila, A., and Serbschi, C., *Sci. Rep.*, 2019, vol. 9, no. 1, p. 14278.
16. Filimonova, M., Shitova, A., Soldatova, O., Shevchenko, L., Saburova, A., Podosinnikova, T., Surinova, V., Shegay, P., Kaprin, A., Ivanov, S., and Filimonov, A., *Int. J. Mol. Sci.*, 2022, vol. 23, p. 730.

Translated by M. Batrukova

Publisher's Note. Pleiades Publishing remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.